DIGITAL IMAGE PROCESSING

Lecture 6

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Chapter 3:
Fundamentals of Spatial Filtering

Fundamentals of Spatial Filtering

- Filtering refers to accepting(passing) or rejecting certain frequency components. This effectively smoothens or sharpens the image.

- E.g. Low pass filter, high pass filter, etc.

- Such operations can be directly carried out on image in spatial domain also by using spatial filters (kernels, spatial masks, templates, & windows).

- Spatial filters are more versatile as they are used in linear as well as non-linear filtering (Difficult in frequency domain).

Fundamentals of Spatial Filtering

- Types of Spatial Filtering
- 1) Point to point (pixel to pixel) operation (discussed so far)
- 2) Mask based (Neighborhood) operations

    i) Operation with 3x3 filter (E.g. Mean, max, min, etc)

    ii) Correlation or Convolution

- Linear vs Non-Linear Filter

  If the operation performed on the image pixels is linear, then the filter is called a linear spatial filter, otherwise nonlinear.

Operation with 3x3 Filter

- At any point (x, y) in the image, the response, g(x, y), of the filter is the sum of products of the filter coefficients and the image pixels encompassed by the filter:

$$
\begin{aligned}
g(x, y) = {} & f(x-1,y-1).w1 + f(x-1,y).w2 + f(x-1,y+1).w3 \\
& + f(x,y-1).w4 + f(x, y).w5 + f(x,y+1).w6 \\
& + f(x+1,y-1).w7 + f(x+1,y).w8 + f(x+1,y+1).w9
\end{aligned}
$$

- For the mask of size m x n, we assume
  $m = 2a + 1$;
  $n = 2b + 1$;
  where a & b are positive integers.

- 3x3 is the smallest filter.

Operation with 3x3 Filter

- Generalized equation:

- $$g(x, y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s, t)\, f(x + s, y + t)$$

where,
x & y are varied so that each pixel in w visits every pixel in f.

Operation with 3x3 Filter

- 3 x 3 Neighborhood / Mask / Window / Template:

Y

(x - 1)	w(-1,-1) f(x-1, y-1)	w(-1,0) f(x-1, y)	w(-1,1) f(x-1, y+1)
x	w(0,-1) f(x, y-1)	w(0,0) f(x, y)	w(0,1) f(x, y+1)
(x + 1)	w(1,-1) f(x+1, y-1)	w(1,0) f(x+1, y)	w(1,1) f(x+1, y+1)

X

Correlation & Convolution

- Correlation & Convolution are two closely related concepts used in linear spatial filtering.
- *Correlation*: It is a process of moving a filter mask over an image & computing the sum of products at each location.
- *Convolution*: Here, the mechanics are same, except that the filter is first rotated by 180˚.
- Correlation & Convolution are function of displacement. Correlation & Convolution are exactly same if the filter mask is symmetric.
- 1D correlation and convolution of a filter with a discrete unit impulse is shown below.

Origin $f(x, y)$

0	0	0	0	0
0	0	0	0	0
0	0	1	0	0
0	0	0	0	0
0	0	0	0	0

$w(x, y)$

1	2	3
4	5	6
7	8	9

(a)

Padded f

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

(b)

Initial position for w

1	2	3	0	0	0	0	0	0
4	5	6	0	0	0	0	0	0
7	8	9	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

(c)

Full correlation result

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	9	8	7	0	0	0
0	0	0	6	5	4	0	0	0
0	0	0	3	2	1	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

(d)

Cropped correlation result

0	0	0	0	0
0	9	8	7	0
0	6	5	4	0
0	3	2	1	0
0	0	0	0	0

(e)

Rotated w

9	8	7	0	0	0	0	0	0
6	5	4	0	0	0	0	0	0
3	2	1	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

(f)

Full convolution result

0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	1	2	3	0	0	0
0	0	0	4	5	6	0	0	0
0	0	0	7	8	9	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0

(g)

Cropped convolution result

0	0	0	0	0
0	1	2	3	0
0	4	5	6	0
0	7	8	9	0
0	0	0	0	0

(h)

Correlation & Convolution

- Summarizing in equation form, we have that

- The Correlation of a filter w(x, y) of size mxn with an image f(x, y)is given by:

$$w(x, y) \star f(x, y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s, t) f(x + s, y + t)$$

- The Convolution of w(x, y) and f(x, y) is given by:

$$w(x, y) \bigstar f(x, y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s, t) f(x - s, y - t)$$

Vector representation of Linear Filtering

w_1	w_2	w_3
w_4	w_5	w_6
w_7	w_8	w_9

$$R = w_1 z_1 + w_2 z_2 + \ldots + w_9 z_9$$

$$= \sum_{k=1}^{9} w_k z_k$$

$$= w^T z$$

Where, w & z are 9-dimensional vectors formed from coefficients of the mask & image intensities encompassed by the mask, resp.

Generating Spatial Filter Masks

1) Average Mean Filter

- The average value at any location (x, y) in the image is the sum of the nine intensity values in the 3 x 3 neighborhood centered on (x, y) divided by 9.
- If z_i, i = 1, 2, …, 9 denote these intensities, then the average is:

$$R = \frac{1}{9}\sum_{i=1}^{9} z_i$$

$$\frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \qquad \frac{1}{16} \times \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Generating Spatial Filter Masks

General implementation for filtering an M x N image with a weighted average filter of size m x n is given by:

$$g(x, y) = \frac{\sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t) f(x+s, y+t)}{\sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t)}$$

2) Exponential Filter

- Some applications have a continuous function of 2 variables. E.g. Gaussian function Spatial filter mask has the basic form:

$$h(x, y) = e^{-\frac{x^2+y^2}{2\sigma^2}}$$

where, σ is standard deviation

Generating Spatial Filter Masks

Ex. 1) 8x8 Pseudo image with a single edge (High Frequency) of 10 & 50. Remove using a 3x3 size averaging mask.

10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50

8x8 Image

Generating Spatial Filter Masks

10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50

1	1	1	1
-----	1	1	1
9	1	1	1

Generating Spatial Filter Masks

0	0	0						
0	10	10	10	10	10	10	10	10
0	10	10	10	10	10	10	10	10
	10	10	10	10	10	10	10	10
	10	10	10	10	10	10	10	10
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50

1	1	1	1
-----	1	1	1
9	1	1	1

Generating Spatial Filter Masks

0	0	0						
0	4.44	10	10	10	10	10	10	10
0	10	10	10	10	10	10	10	10
	10	10	10	10	10	10	10	10
	10	10	10	10	10	10	10	10
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50

1	0	0	0
-----	0	10	10
9	0	10	10

Generating Spatial Filter Masks

0	0	0	0					
0	4.44	6.66	10	10	10	10	10	10
0	10	10	10	10	10	10	10	10
	10	10	10	10	10	10	10	10
	10	10	10	10	10	10	10	10
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50

1	0	0	0
-----	10	10	10
9	10	10	10

Generating Spatial Filter Masks

0		0	0	0				
0	4.44	6.66	6.66	10	10	10	10	10
0	10	10	10	10	10	10	10	10
	10	10	10	10	10	10	10	10
	10	10	10	10	10	10	10	10
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50

1	0	0	0
-----	10	10	10
9	10	10	10

Generating Spatial Filter Masks

0			0	0	0			
0	4.44	6.66	6.66	6.66	10	10	10	10
0	10	10	10	10	10	10	10	10
	10	10	10	10	10	10	10	10
	10	10	10	10	10	10	10	10
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50

1	0	0	0
-----	10	10	10
9	10	10	10

Generating Spatial Filter Masks

0	0	0	0	0	0	0	0	0	
0	4.44	6.66	6.66	6.66	6.66	6.66	6.66	4.44	0
0	6.66	10	10	10	10	10	10	10	0
	10	10	10	10	10	10	10	10	
	10	10	10	10	10	10	10	10	
	50	50	50	50	50	50	50	50	
	50	50	50	50	50	50	50	50	
	50	50	50	50	50	50	50	50	
	50	50	50	50	50	50	50	50	

$$\frac{1}{9}\begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Generating Spatial Filter Masks

0 0 0 0 0 0 0 0 0

	4.44	6.66	6.66	6.66	6.66	6.66	6.66	4.44	0
	6.66	10	10	10	10	10	10	6.66	0
0	6.66	10	10	10	10	10	10	6.66	0
0	15.55	10	10	10	10	10	10	10	0
0	50	50	50	50	50	50	50	50	0
	50	50	50	50	50	50	50	50	
	50	50	50	50	50	50	50	50	
	50	50	50	50	50	50	50	50	

1	0	10	10
-----	0	10	10
9	0	50	50

Generating Spatial Filter Masks

0	0	0	0	0	0	0	0	0	
0	4.44	6.66	6.66	6.66	6.66	6.66	6.66	4.44	0
0	6.66	10	10	10	10	10	10	6.66	0
0	6.66	10	10	10	10	10	10	6.66	0
0	15.55	23.33	10	10	10	10	10	10	0
0									0
	50	50	50	50	50	50	50	50	
	50	50	50	50	50	50	50	50	
	50	50	50	50	50	50	50	50	
	50	50	50	50	50	50	50	50	

$$\frac{1}{9} \begin{array}{|c|c|c|} \hline 10 & 10 & 10 \\ \hline 10 & 10 & 10 \\ \hline 50 & 50 & 50 \\ \hline \end{array}$$

Generating Spatial Filter Masks

0	0	0	0	0	0	0	0	0
0	4.44	6.66	6.66	6.66	6.66	6.66	6.66	4.44
0	6.66	10	10	10	10	10	10	6.66
0	6.66	10	10	10	10	10	10	6.66
0	15.55	23.33	23.332	3.33	23.33	23.33	23.33	15.55
0	24.44	36.66	50	50	50	50	50	50
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50

1	10	10	10
-----	50	50	50
9	50	50	50

Generating Spatial Filter Masks

0	0	0	0	0	0	0	0	0
0	4.44	6.66	6.66	6.66	6.66	6.66	6.66	4.44
0	6.66	10	10	10	10	10	10	6.66
0	6.66	10	10	10	10	10	10	6.66
0	15.55	23.33	23.332	3.33	23.33	23.33	23.33	15.55
0	24.44	36.66	36.66	36.6 6	36.66	36.66	36.66	24.44
0	33.33	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50
	50	50	50	50	50	50	50	50

1	50	50	50
-----	50	50	50
9	50	50	50

Generating Spatial Filter Masks

0 0 0 0 0 0 0 0 0

0	4.44	6.66	6.66	6.66	6.66	6.66	6.66	4.44	0
0	6.66	10	10	10	10	10	10	6.66	0
0	6.66	10	10	10	10	10	10	6.66	0
0	15.55	23.33	23.332	3.33	23.33	23.33	23.33	15.55	0
0	24.44	36.66	36.66	36.66	36.66	36.66	36.66	24.44	0
0	33.33	50	50	50	50	50	50	33.33	
0	33.33	50	50	[illegible]	50	50	50	33.33	0
0	22.22	33.33	33.33	[illegible]	.33	33.33	33.33	22.22	0
							0	0	0

50	50	0
50	50	0
0	0	0

Generating Spatial Filter Masks

10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	23.33	23.33	23.33	23.33	23.33	23.33	10
50	36.66	36.66	36.66	36.66	36.66	36.66	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50
50	50				50	50	50

$\frac{1}{9}$

1	1	1
1	1	1
1	1	1

Generating Spatial Filter Masks

- In the resultant image the Low frequency region has remained unchanged.
- Sharp transition between 10 & 50 has changed from 10 to 23.33 to 36.66 and finally to 50.
- Thus, Sharp edges has become blurred.
- Best result when used over image corrupted by Gaussian noise.
- Other types of low pass averaging mask are:

$$\frac{1}{6}\begin{bmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix} \qquad \frac{1}{10}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Order-Statistic Filters

- These are non-linear spatial filters whose response is based on ordering (increasing / decreasing) the pixels contained in the image area encompassed by the filter.
- Then replacing the value of the center with the middle value determined by ranking result.
- E.g. Median filter, Max filter, Min Filter

- i) *Median Filter*:
- Popular with certain random noise and impulse noise (Salt & Pepper noise).
  - They provide excellent noise reduction
  - Comparatively less blurring than linear smoothing filter of same size.

Order-Statistic Filters

Ex. 2) 8x8 Pseudo image with a single edge (High Frequency) of 10 & 50. Remove using a 3x3 size median filter mask.

10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	250	10	10	10	10	10	10
10	10	10	10	10	10	10	10
50	50	50	50	250	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50

8x8 Image

Order-Statistic Filters

10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	250	10	10	10	10	10	10
10	10	10	10	10	10	10	10
50	50	50	50	250	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50

8x8 Image with blank mask

Order-Statistic Filters

10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	250	10	10	10	10	10	10
10	10	10	10	10	10	10	10
50	50	50	50	250	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50

8x8 Image with blank mask

Order-Statistic Filters

10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
50	50	50	50	250	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50

8x8 Image with blank mask

Order-Statistic Filters

10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
50	50	50	50	250	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50

8x8 Image with blank mask

Order-Statistic Filters

10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
50	50	50	50	250	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50

8x8 Image with blank mask

Order-Statistic Filters

10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
10	10	10	10	10	10	10	10
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50
50	50	50	50	50	50	50	50

8x8 Image with blank mask

Order-Statistic Filters

- ii) *Max Filter*:
- It is used to find the **brightest points** in an image.
  - Response of a 3 x 3 max filter is given by

$$R = \max\{z_k \mid k = 1, 2, \ldots, 9\}$$

- iii) *Min Filter*:
  - Used to find the **darkest points** in an image.

Sharpening Spatial Filter

- The principal objective of sharpening is to highlight transitions in intensity.
- Applications: Electronics Printing, Medical Imaging, Industrial Inspection, Autonomous guidance in military systems, etc.
- Derivatives of a digital function are defined in terms of differences:

- *First Derivative*:

i) must be zero in areas of constant intensity.

ii) must be nonzero at the onset of an intensity step / ramp.

iii) Must be nonzero along ramps.

- First order derivat ... ifference:

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

Sharpening Spatial Filter

- Second derivative:

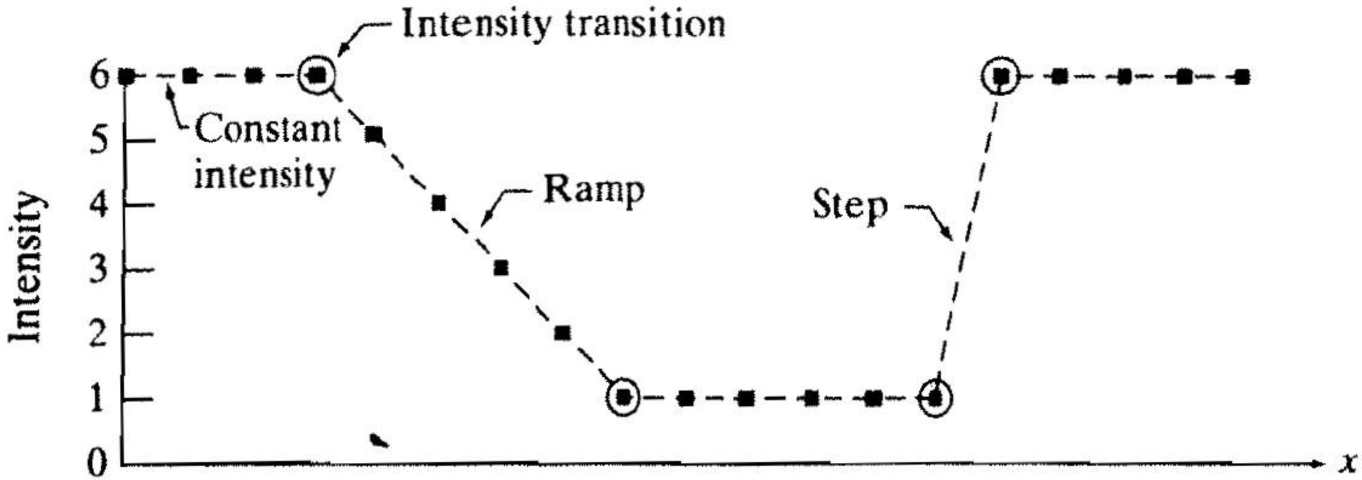
i) must be zero in constant areas

ii) Must be nonzero at the onset and end of an intensity step / ramp

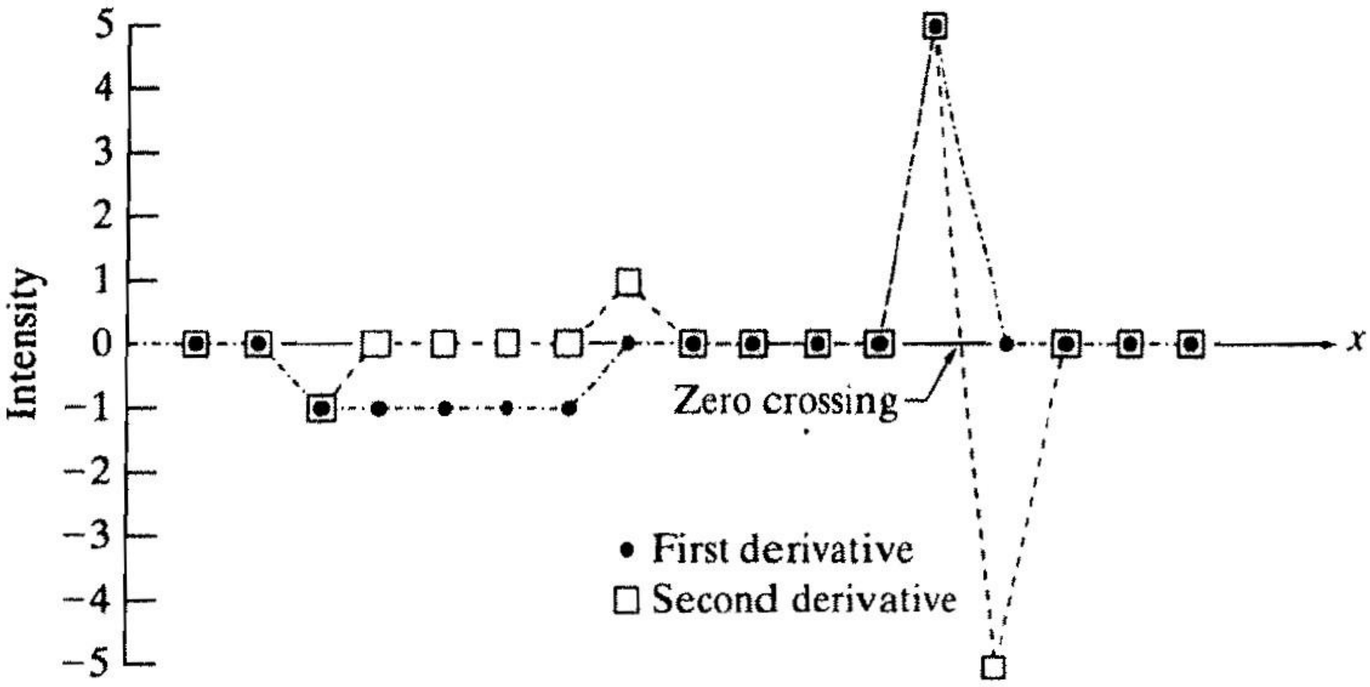
iii) Must be zero along ramps of constant slope.

- Second order derivative of f(x) as the difference:

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$



Scan line	6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6 → x
1st derivative		0	0	−1	−1	−1	−1	−1	0	0	0	0	0	5	0	0	0	0	
2nd derivative		0	0	−1	0	0	0	0	1	0	0	0	0	5	−5	0	0	0	



The Laplacian

- Isotropic Filter: They are rotation invariant.

- Laplacian is simplest Isotropic derivative operator. It is defined as:

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

- Laplacian is a linear operator.

The Laplacian

- To express the equation in discrete form,
- In x-direction,

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

- In y-direction,

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

- Thus, discrete Laplacian of 2 variables is,

$$\nabla^2 f(x, y) = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

The Laplacian

- Filters to implement Laplacian:

0	1	0
1	−4	1
0	1	0

1	1	1
1	−8	1
1	1	1

0	−1	0
−1	4	−1
0	−1	0

−1	−1	−1
−1	8	−1
−1	−1	−1

The Laplacian

- If background features need to be recovered by still preserving the sharpening effect of the Laplacian, then
- The basic way to use the Laplacian for image sharpening is:

$$g(x, y) = f(x, y) + c\left[\nabla^2 f(x, y)\right]$$

Where,

f(x, y) – input image

g(x, y) – sharpened image

c – (-1) or (+1) for filters with negative & positive center resp.

Unsharp Masking & Highboost Filtering

- The process of subtracting an unsharp (smoothed) version of an image from the original image is called unsharp masking.

- It consists of three steps:
- i) Blur the original image.

$$\bar{f}(x, y)$$

- ii) Subtract the blurred image from the original (results in mask).

$$g_{mask}(x, y) = f(x, y) - \bar{f}(x, y)$$

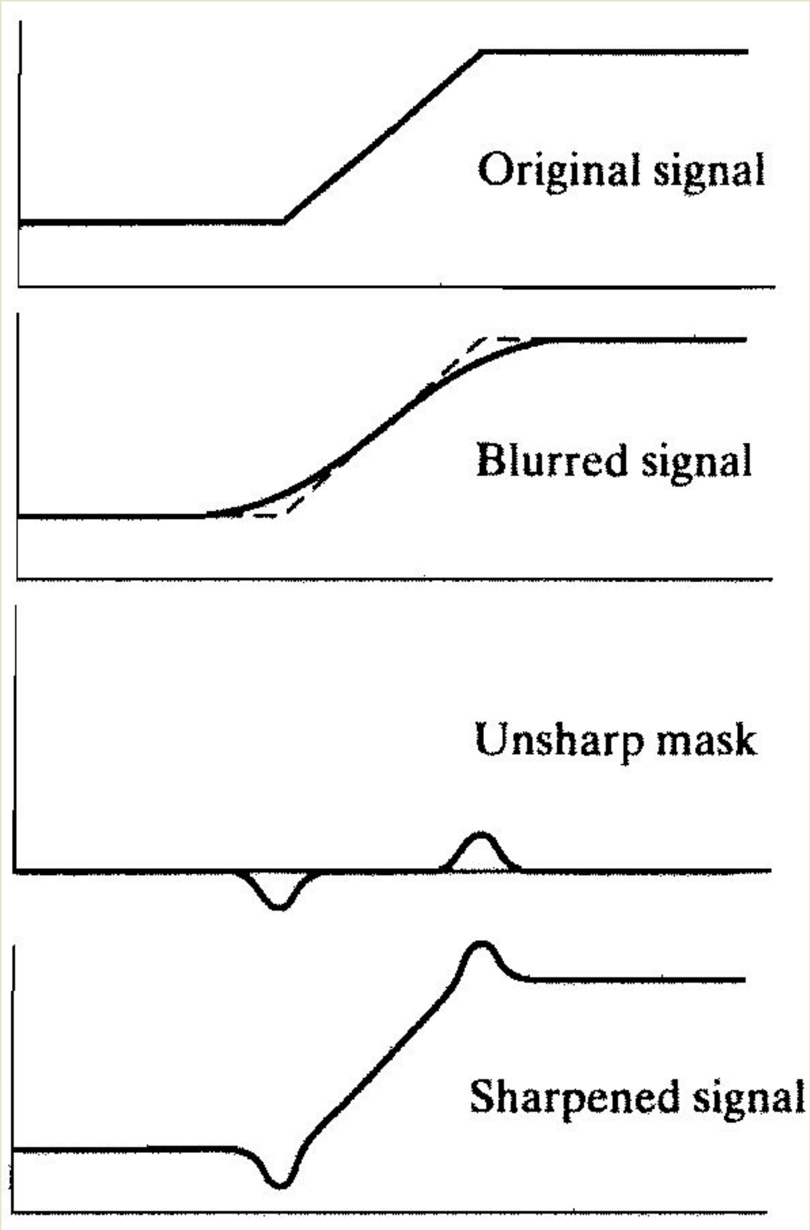
- iii) Add the mask to the original.

$$g(x, y) = f(x, y) + k * g_{mask}(x, y)$$

where if, k = 1, unsharp masking, k > 1, highboost filtering, k < 1, de-emphasizes the contribution of unsharp mask.

Unsharp Masking & Highboost Filtering

- Mechanics of unsharp masking



Using First-Order Derivative (Gradient)

- First derivative are implemented using the magnitude of the gradient.
- For image f(x, y), the gradient of f at (x, y) is given by:

$$\nabla f \equiv \text{grad}(f) \equiv \begin{bmatrix} g_x \\ g_y \end{bmatrix} = \begin{bmatrix} \dfrac{\partial f}{\partial x} \\ \dfrac{\partial f}{\partial y} \end{bmatrix}$$

- This vector points in the direction of the greatest rate of change of f at (x, y).
- Its magnitude is given by:

$$M(x, y) = \text{mag}(\nabla f) = \sqrt{g_x^2 + g_y^2}$$

- Sometimes

$$M(x, y) \approx |g_x| + |g_y|$$

Using First-Order Derivative (Gradient)

z_1	z_2	z_3
z_4	z_5	z_6
z_7	z_8	z_9

- Roberts operators

−1	0
0	1

0	−1
1	0

- Sobel operators

−1	−2	−1
0	0	0
1	2	1

−1	0	1
−2	0	2
−1	0	1

Acknowledgment

- "Digital Image Processing", 2/ E, Rafael C. Gonzalez & Richard E. Woods,

Questions?